

Column Calculations

$$F_y = 500 \text{ MPa}$$

$$F_{ck} = 25 \text{ MPa}$$

$$\text{Actual Required} = 3941 \text{ mm}^2$$

$$A_s(\%) = 0.62$$

$$\text{Box size} = 16, \quad \text{Bars No.} = 20$$

$$P_u = 5493.41 \text{ kN}, \quad M_z = 52.72 \text{ kN-m}, \quad M_y = 68.88 \text{ kN-m}$$

$$P_{uz} = 0.45f_{ck}A_c + 0.75f_yA_{sc}$$

$$A_c = (800 \times 800) - 3941$$

$$P_{uz} = 8633.539 \text{ kN}$$

$$\frac{P_u}{P_{uz}} = \frac{5493.41}{8633.585} = 0.636 > 0.2$$

$$d' = \frac{n-2}{2} \times \left(1 + \frac{5}{10 \times 1.05}\right) \\ = 1.72$$

$$D = 800 \text{ mm}$$

$$\text{min clear cover} = 40 \text{ mm}$$

$$d' = 40 + 1 \times \left[\frac{16}{2}\right]$$

$$d' = 48 \text{ mm}$$

$$\frac{d'}{D} = \frac{48}{800} = 0.06$$

$$e_{min} = \frac{L_0}{500} + \frac{D}{30}$$

$$L = 3100 \text{ mm}$$

$$e_{min} = \frac{3100}{500} + \frac{800}{30} = 32.87 \text{ mm}$$

$$\frac{P_u}{f_{ck}bD} = \frac{0.62}{25} = 0.0248$$

$$\frac{P_u}{f_{ck}bD} = \frac{5493.11 \times 10^3}{3 \times 10^6 \times 800 \times 10^{-3}} = 0.286$$

$$\frac{M_u}{f_{ck}bD^2} = 0.067 \quad (\text{From Interaction diagram})$$

$$M_{uz} = 0.069 \times 25 \times (800)^2 \times 10^6 \times 10^{-9}$$

$$M_{uz} = 857.6 \text{ kN-m}$$

$$M_{uy} = 857.6 \text{ kN-m}$$

$$\begin{aligned} \left(\frac{M_{uy}}{M_{uy}} \right)^{1.92} + \left(\frac{M_{uz}}{M_{uz}} \right)^{1.92} &= \left(\frac{68.88}{857.6} \right)^{1.92} + \left(\frac{52.72}{857.6} \right)^{1.92} \\ &= 0.021 \leq 1 \quad \Rightarrow \quad \text{Safe} \end{aligned}$$